TEXORA · LECTURE SHEET

# Project Plan for a New Dyeing and Finishing Facility

*This closing topic assembles the whole course: capacity arithmetic from Topics 6–10, water and steam from Topics 2 and 11, and layout logic, into the plan an engineer would put in front of an investor — sized, balanced and costed in structure.*

WP-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 12 · SPECIALIZATION — WET PROCESSING

## 1. Start from the product programme, not from machines

A facility is designed backwards from a **product programme**: what goods, how much, of what quality, for which market. The programme fixes the process route, and the route fixes the machines. A plan that begins with a machinery catalogue produces a factory that can make whatever its machines allow — rarely what the market ordered. The programme states: fibre(s) and fabric forms (here: cotton and cotton-blend knits), capacity (here: 10,000 kg/day of dyed and finished fabric), quality level (export knitwear), and working pattern (20 h/day production, 300 days/year — typical planning values, adjusted per project).

## 2. Capacity sizing — the course arithmetic, applied

**WORKED EXAMPLE 1 — SIZING THE DYEING SECTION FOR 10,000 KG/DAY**

Use the production unit of Topic 7: a 4-tube jet, 250 kg per tube = 1,000 kg per batch, standard cycle 6.5 h.

Across a real shade programme the average cycle lengthens — dark shades with long wash-off dominate knit dyeing — so plan on a programme-average cycle of about 9 h: ⌊20/9⌋ = 2 batches = 2,000 kg/day, discounted for changeovers, cleaning and maintenance to an effective output of **1,500 kg per machine per day** (stated as a planning assumption so the investor sees it).

Machines required = 10,000 / 1,500 = 6.67 → **7 dyeing machines** (the seventh also provides cover during maintenance).

Preparation and finishing must match 10,000 kg/day: soft-flow preparation in the jets themselves for knits (prepare–dye in one machine is standard knit practice), hydro-extractors and stenters sized by Topic 9 arithmetic — each stenter at ~700 kg/h covers the daily tonnage in about 15–16

running hours, so **2 stenters** give capacity and cover. Compactors follow the same loads-per-day logic as Topic 10.

The pattern is universal: define the production unit, prove its effective daily output with the floor function and a stated utilisation, divide the programme by it, round **up**, and check every other section against the same daily tonnage so no section starves or floods another.

## 3. Water, steam and effluent — the utility sizing

**WORKED EXAMPLE 2 — UTILITIES FOR 10,000 KG/DAY**

**Water:** total process water (preparation, dyeing, wash-off, rinsing, finishing) at a planning figure of 100 L per kg of fabric → 10,000 × 100 = 1,000,000 L/day = **1,000 m³/day**. Source, storage (at least one day), and softening capacity are sized to this figure.

**Steam:** at a planning figure of 4.5 kg steam per kg of fabric → 45,000 kg/day. Over 20 working hours = **2,250 kg/h** average demand; choose boiler capacity above peak demand — a **3 t/h** boiler (or 2 × 2 t/h for cover) is the planning answer.

**Effluent:** essentially all process water leaves as effluent: the ETP must treat ≈ 1,000 m³/day → at 24 h operation ≈ 42 m³/h; design at **45–50 m³/h** with equalisation. ETP capacity is not an afterthought: in most jurisdictions the plant's consent to operate is written around it.

Planning figures are labelled as such; a real project replaces them with figures from the selected machines' verified consumption data.

## 4. Layout — material flows downhill through the plan

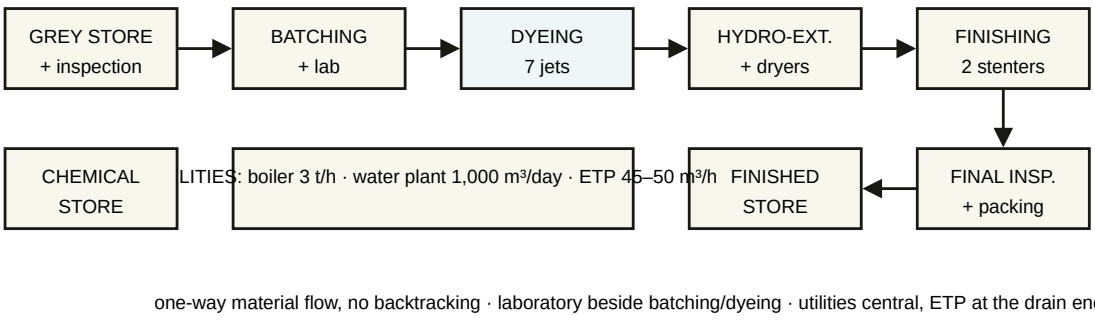



Fig. 1 — Block layout of the planned 10 t/day knit dyeing and finishing facility (schematic, not to scale). Every number in the boxes comes from Worked Examples 1 and 2.

Layout rules the block plan obeys: material moves in one direction without crossing its own path; wet and dry zones are separated; the laboratory sits between batching and dyeing (its two customers); chemical storage is bunded and close to, but not inside, the dyeing hall; utilities are central to shorten steam and water runs; the ETP sits at the low end of the site drainage. Building services — effluent drains with falls, steam mains with condensate return, ventilation over hot machines — are drawn at this stage, not discovered during installation.

## 5. Structure of the project document

| SECTION | CONTENT |
|---|---|
| Product programme and market | Goods, capacity, quality, buyers' compliance requirements (effluent and chemical standards) |
| Process route and machinery list | Section-by-section machine schedule with the capacity proof of Section 2 for each |
| Utilities and environment | Water, steam, power, compressed air; ETP design and discharge consent route |
| Layout and building | Block plan (Fig. 1) developed to machine footprints, services and expansion space |
| Manpower and organisation | Shifts, skills, laboratory and quality staffing |
| Capital and running cost structure | Machinery, building, utilities, installation; unit processing cost per kg built from water/steam/chemical/labour rates — the structure is presented, with figures filled from quotations at project stage |
| Implementation schedule | Civil work, machine delivery and installation, commissioning, trial production — phased so utilities and ETP are ready before the first dyeing machine |

## 6. Glossary — key terms

- **Product programme:** the defined range and volume of goods the facility is designed to make.
- **Utilisation:** effective output as a fraction of arithmetic maximum, after changeovers and losses.
- **ETP:** effluent treatment plant.
- **Consent to operate:** the regulatory permission tying production to environmental compliance.

## 7. Key points

- Design backwards: programme → route → machine units → counts (10,000 ÷ 1,500 = 6.67 → 7 jets).
- Utilities are sized from the same tonnage: 1,000 m³/day of water, a 3 t/h boiler, a 45–50 m³/h ETP.
- Every planning figure in a project document is labelled with its basis — machine data, planning norm or assumption — so the investor can audit the arithmetic.

## 8. Summary

The project plan is the course in one document: recipes scaled to tonnes, machines counted by cycle arithmetic, water and steam carried through every section, and a layout that lets material flow one way from grey store to finished store. An engineer who can build this plan — and defend every number in it — can build the factory.

## 9. Practice

1. Re-size the dyeing section for a programme of 6,000 kg/day using the same machine unit and effective output as Worked Example 1. How many machines, and what total installed batch capacity does that give per day at 3 arithmetic batches?
2. For the 6,000 kg/day plant, estimate daily water demand and average steam demand (20 h day) using the planning figures of Worked Example 2, and propose a boiler size.
3. In Fig. 1, why is the laboratory placed beside batching and dyeing rather than beside final inspection?
4. Name two planning assumptions in this sheet that a real project must replace with verified data before money is committed.

### Answer key

1. Machines = 6,000/1,500 = 4.0 → 4 machines. Installed arithmetic capacity = 4 machines × 1,000 kg × 3 batches = 12,000 kg/day (the programme uses half of it effectively — utilisation headroom for peaks and maintenance).
2. Water = 6,000 × 100 L = 600 m³/day. Steam = 6,000 × 4.5 = 27,000 kg/day → 1,350 kg/h average → a 2 t/h boiler gives working margin.
3. The laboratory's production customers are the dyehouse decisions made before and during dyeing — lab dips, recipe issue, shade corrections. Beside batching/dyeing its turnaround is minutes; final inspection needs only its test reports, which travel on paper or network, unlike samples.
4. Any two of: the 1,500 kg/day effective machine output (utilisation assumption); the 100 L/kg water figure; the 4.5 kg steam/kg figure; the 300-day/20-hour working pattern; machine prices and utility tariffs in costing.